

Crossfade

A capability-aware evidence and experiment platform for wave-propagation inference

Full thesis, specification v0.3 and build pack v0.3, in one document

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Abstract

Crossfade connects sonar, HF skywave radar and RF systems through a common channel vocabulary anchored on the channel spreading function, and it measures where learned structure transfers between domains instead of assuming it. This document is the complete thesis: the motivation and the three separable contracts the platform commits to; the physics and mathematics it relies on, with the regime coordinates that turn “does it transfer?” into a preregisterable hypothesis; the identifiability limits that no estimator removes; the architecture, the five record types and the twenty-five invariants that make every result auditable; the models, the training objective and the baseline ladder; the typed treatment of uncertainty; the evaluation protocol with its three gates; the build plan in four slices with their acceptance criteria; the decision register; and the open questions for the sonar lead, two of which are on the critical path. Every requirement is tagged as an engineering guarantee [G], a provisional choice [P], or a hypothesis [H], and the tag decides what happens when it fails: a bug is fixed, a component is swapped, or a result is recorded.

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Part I

Motivation and claims

1 What Crossfade is

The idea came from a conversation between a sonar and cross-domain signals lead and an ML lead. Every display anyone looks at, a range-Doppler map, a beam-level spectrogram, a cepstrum, is a measurement of the propagation channel’s response, and the time-varying impulse response or the scattering function is what actually carries the answer: whether a target is there, whether it is closing, what the environment is doing. If you can estimate it, in the sonar lead’s words, you have won. The same object describes an ocean waveguide, the Earth-ionosphere duct and an urban RF channel; the ionosphere is “the ocean flipped upside down”. The question that followed was whether a model could learn what those channels have in common so that each new domain, environment or system needs less bespoke data and modelling, and where the modern tools, self-supervised encoders, physics-informed networks, large language models, actually fit.

Crossfade is the version of that idea that survived two rounds of review. The channel is still the centre. What changed is that we now know which parts of the idea the physics supports, which parts are hypotheses to measure, and where the machine learning and the language model sit so that they cannot quietly take over the science.

The one-sentence design. Every quantity the platform estimates is defined as a functional of a named channel representation; every record carries units, coordinates, capabilities, processing history and lineage; every run has typed result states that are never filled with a passing value by default; every claim of transfer resolves to a preregistered comparison at a declared physical regime; and the language model reads records and writes drafts but never a number.

2 Three separable contracts

Crossfade commits to three contracts, and the platform stays useful if only the first holds.

Contract	Claim	Evidence that it holds
Interoperability	Any wave-sensing system’s outputs can be ingested with units, coordinates, processing history and capability description intact, and replayed	A new domain connects through a domain pack without changing the core; a result reproduces from archived evidence
Representation transfer	A shared encoder pretrained on one domain reduces the target-domain data needed to estimate a declared quantity	Frozen-encoder few-shot gain over from-scratch at declared budgets, with effect size and uncertainty
Evidence fusion	Multiple results about one situation combine under a declared dependence model	Duplicate evidence is never double-counted; unknown dependence withholds numerical fusion

3 Three kinds of statement

Tag	Meaning	On failure
[G] Guarantee	A property the software enforces: immutability, provenance, typed states, leakage checks, reproducible comparison	Bug; fix the software; a test exists or must be written
[P] Provisional	An architectural choice made now and replaceable later: canonical anchor kind, encoder family, storage backend	Swap the component; contracts unchanged
[H] Hypothesis	A scientific claim to be measured: transfer at matched regime, usefulness of a regime coordinate, benefit of routing	Record the result in the atlas; the platform is unaffected

4 What Crossfade is not

- Not a universal physics-informed neural network. PDE-residual PINNs fail at high wavenumber through spectral bias [19], which is exactly the regime of wave propagation. Physics enters as structural priors on the channel and as forward models in domain packs, where available.
- Not an LLM that reads latent vectors. The LLM sits on typed records, explains, retrieves, and drafts pack metadata for human approval.
- Not a replacement for existing processing chains. Legacy systems connect through output-only packs and are never written to.
- Not a single latent space, and not a required reconstruction of one universal object from every input. Sharing is partial, per descriptor, and earned by benchmark.

Two tensions in the original conversation are resolved as follows. First, “factor out the channel” versus “the channel is the information”: a variable is a nuisance relative to a task, never inherently, so each task specification says what is retained, conditioned on, or marginalized. Second, “domain-invariant” versus “no information loss”: these cannot both hold in one compressed representation, so the archive preserves the source evidence, each derived representation declares what it retains, and private pathways carry what the shared one drops.

5 How the thesis was formed

The record of how we got here is kept in the repository’s **archive/** directory. In order:

1. **The transcript.** The sonar lead’s claims: the channel as the common thread; cepstral and blind channel-estimation work; the overlapped-source problem where many signals share one band and multipath traces cannot be assigned to sources; the practical question of window and FFT lengths across domains; the desire for a training objective that aligns the spaces “naturally without carving something up”.
2. **Two plans.** A first-principles plan that wrote down the generative structure $y_d = \mathcal{M}_d[\mathcal{F}_d(\theta; c_d)] + \varepsilon_d$ and derived the architecture from it, and a physics-modelling plan that supplied the representation atlas, the capability registry, the evidence graph and a workbench that stays useful when every transfer hypothesis fails.
3. **The synthesis.** A referee document found the plans agree on roughly eighty percent of their content and are complementary where they differ, and added three things neither said: name the anchor as the spreading function; make the regime a dimensionless coordinate; take self-supervised pretraining on paired views seriously.
4. **Spec v0.1 and an external architecture review.** The review’s facts were examined claim by claim and found correct; its scaffolding was about twice the needed size (thirteen

- record types, twelve slices, a **conditional** assessment state with an authorization step). Its contract amendments were adopted with three modifications; its scaffolding was collapsed.
5. **Spec v0.2, the build-pack assessment, and v0.3.** The v0.2 build pack was dissected against one question: is this the simplest way to achieve the same result? The answer produced build pack v0.3, one Python package with five modules, and spec v0.3 aligned to it.
 6. **The W0 workbench.** The first slice was implemented test-first on a branch and is waiting on the sonar lead’s review before merging.

Corrections we made to our own earlier claims. These are worth listing because the reviewer for whom this document is written would have caught each of them.

- v0.1 called delay–Doppler “the” narrowband form and delay–scale “the” wideband form. Both always exist; what changes with $TB(v/c)$ is sparsity (Proposition 7.2).
- v0.1 called resampling, windowing and carrier retune “conventions”. They are lossy processing or physical intervention, and each transform is now classified and tested individually.
- v0.1 claimed that contrasting a waveform with its own spectrogram and its own estimated CIR makes shared content identifiable. A deterministic re-representation has no nuisance to strip; only views with independent nuisance variation do (Section 11).
- v0.1’s placeholder regime ranges were disjoint in M and β_B while the pilot required overlap. Resolved as per-descriptor matching, which produced the overlap result (Proposition 9.1).
- Citation corrections found while checking sources against the record: the bilinear-model paper is by Tian, Lee, Romberg, Durofchalk and Sabra (2020), not Sabra and Dowling; the ray-based blind deconvolution paper is Sabra, Song and Dowling (2010); Altes’ wideband ambiguity paper is 1973; the 2011 time-scale characterization paper is by Chude-Okonkwo, Ngah and Abd Rahman; the CSI-CLIP++ authors are Jiang, Yu, Li, Gao and Xu.

Part II

Physics and mathematics

6 Notation and conventions

Symbol	Meaning	Units
$x(t), y(t)$	transmitted and received signal (complex baseband unless stated)	–
$s(t)$	source waveform; $s_a(t)$ its analytic signal	–
$h(t, \tau)$	time-varying impulse response (Bello)	s^{-1}
$S_H(\tau, \nu)$	delay–Doppler spreading function	–
$h_W(\tau, \alpha)$	delay–scale spreading function	–
$C_H(\tau, \nu)$	scattering function, $\mathbb{E} S_H ^2$ under WSSUS	–
$P(\tau)$	power–delay profile (PDP)	–
f_c, B, T	carrier, bandwidth, observation length	Hz, Hz, s
c	propagation speed in the medium	m/s
v	radial relative speed, positive when closing	m/s
$M = v/c$	Mach number	1
$\beta_B = B/f_c$	fractional bandwidth	1
σ_τ, σ_ν	RMS delay spread, RMS Doppler spread	s, Hz
$\Pi_\tau, \Pi_\nu, \Pi_L, \Pi_f, \Pi_h, \Pi_s$	regime groups, Section 9	1
$\{(\tau_k, a_k)\}_{k=1}^K$	arrival list: delays and complex gains	s, 1
D, z_s, z_r, r	water depth, source depth, receiver depth, range	m
c_b, ρ_b	bottom half-space sound speed and density	m/s, kg/m ³

Fourier transform convention: $X(f) = \int x(t) e^{-j2\pi f t} dt$. Scale $\alpha > 1$ means time compression (a closing source). Every equation is tagged: **[G]** for something the software enforces, **[P]** for a convention or modelling choice that can be swapped without changing a contract, **[H]** for a scientific claim the benchmark measures. Items marked *Check* are the ones we most want the reader to confirm, correct, or replace.

7 The channel and its representations

7.1 Linear time-varying channel

A single-input single-output propagation channel is modelled as a linear time-varying (LTV) operator $\mathcal{H}$ with kernel $h(t, \tau)$ [2]:

$$y(t) = (\mathcal{H}x)(t) = \int h(t, \tau) x(t - \tau) d\tau. \quad (1)$$

$h(t, \tau)$ is the response at time t to an impulse launched at time $t - \tau$. Its Fourier transform in t is the *delay-Doppler spreading function*,

$$S_H(\tau, \nu) = \int h(t, \tau) e^{-j2\pi \nu t} dt, \quad y(t) = \iint S_H(\tau, \nu) x(t - \tau) e^{j2\pi \nu t} d\tau d\nu, \quad (2)$$

so the channel is a superposition of delay-and-frequency-shift operators. The other two Bello functions follow by Fourier transform in τ : the time-varying transfer function $L_H(t, f) = \int h(t, \tau) e^{-j2\pi f \tau} d\tau$ and the Doppler-variant transfer function.

Proposition 7.1 (Existence). *Equation (2) holds for any LTV operator whose kernel $h(t, \tau)$ is a tempered distribution in t ; the delay-Doppler representation is not a narrowband approximation but a change of coordinates on the kernel [22].*

This is the correction made to v0.1 of the spec, which had called delay-Doppler “the narrowband form”. What depends on the regime is not existence but *sparsity* (Proposition 7.2).

7.2 Delay-scale (wideband) representation

[P]

The same operator can be written as a superposition of delay-and-scale operators,

$$y(t) = \iint h_W(\tau, \alpha) \sqrt{\alpha} x(\alpha(t - \tau)) d\tau d\alpha, \quad \alpha > 0, \quad (3)$$

where h_W is the delay-scale (wideband) spreading function [1, 5]. A single reflector with radial closing speed v produces, in passband,

$$\alpha = \frac{c + v}{c - v} \approx 1 + \frac{2v}{c} \text{ (monostatic, two-way)}, \quad \alpha = 1 + \frac{v}{c} \text{ (one-way, passive)}. \quad (4)$$

Two conventions are pack-declared and recorded in the anchor’s **normalization** field rather than fixed by the core: the $\sqrt{\alpha}$ energy normalization, and whether the scaling is written $x(\alpha(t - \tau))$ as above or $x(\alpha t - \tau)$, which differ by $\tau \mapsto \alpha\tau$.

7.3 When is one representation sparse? The range-walk number

Consider one path with delay τ_0 and scale α_0 , observed over $t \in [0, T]$, acting on a passband signal $x(t) = u(t)e^{j2\pi f_c t}$ with envelope bandwidth B . In baseband the path is the operator

$$u \mapsto \sqrt{\alpha_0} u(\alpha_0(t - \tau_0)) e^{j2\pi \nu_0 t} e^{-j2\pi f_c \alpha_0 \tau_0}, \quad \nu_0 = (\alpha_0 - 1)f_c, \quad (5)$$

whose Bello kernel is $h(t, \tau) = \sqrt{\alpha_0} e^{j2\pi \nu_0 t} e^{-j2\pi f_c \alpha_0 \tau_0} \delta(\tau - \tau(t))$ with the *drifting delay*

$$\tau(t) = \alpha_0 \tau_0 + (1 - \alpha_0) t. \quad (6)$$

Proposition 7.2 (Range walk). *Let g be the delay-resolution kernel of the waveform (width $1/B$) and $\tilde{h} = h *_{\tau} g$ the resolution-limited kernel. Over an observation of length T , the resolution-limited spreading function $\tilde{S}_H(\tau, \nu)$ of the single path (5) has support of extent*

$$\Delta\tau \approx \max\left(\frac{1}{B}, |\alpha_0 - 1|T\right), \quad \Delta\nu \approx \max\left(\frac{1}{T}, |\alpha_0 - 1|B\right), \quad (7)$$

that is, about $\max(1, \Pi_s)$ resolution cells along each axis, where

$$\boxed{\Pi_s = TB|\alpha_0 - 1| = TBM \text{ (one-way)}} \quad (8)$$

is the range-walk number. In the delay-scale plane the same path is a single point (τ_0, α_0) at the resolution of the wideband ambiguity function.

Sketch. Write $\tilde{h}(t, \tau) = \sqrt{\alpha_0} e^{j2\pi\nu_0 t} g(\tau - \tau(t))$ up to a constant phase. Substituting $s = \tau - \tau(t)$ in the Fourier integral over $t \in [0, T]$,

$$\tilde{S}_H(\tau, \nu) = \frac{\sqrt{\alpha_0}}{\alpha_0 - 1} e^{-j2\pi(\nu - \nu_0)t^*(\tau)} G\left(\frac{\nu - \nu_0}{\alpha_0 - 1}\right) \Big|_{s \in \text{window}}, \quad t^*(\tau) = \frac{\tau - \alpha_0\tau_0}{1 - \alpha_0},$$

where G is the transform of g and has width B . The delay support is the set of τ for which $t^*(\tau) \in [0, T]$, of length $|\alpha_0 - 1|T$; the Doppler support is where G is non-negligible, of width $|\alpha_0 - 1|B$. When either is below one resolution cell the window and kernel widths take over. $\square$

Remark 7.3. Equation (8) is the Kelly–Wishner narrowband condition $TB(2v/c) \ll 1$ for the monostatic case [17], written as a dimensionless group. In the regime coordinates of Section 9, $\Pi_s = M \cdot TB = M\beta_B(f_c T)$. It is a property of the waveform and the motion jointly, so the threshold θ at which a pack switches canonical kind from `delay_doppler` to `delay_scale` is declared and tested by the pack, per regime box, and is not core logic.

Check 1. Is $\Pi_s = TB|\alpha - 1|$ the quantity you would use to decide between delay–Doppler and delay–scale processing, and is a threshold of order one (we have used $\theta \approx 0.1$ as a placeholder) the right order? For the passive wideband sonar case, is there structure in the modal picture that the delay–scale form loses at ranges you care about?

7.4 Realization versus statistic

[G] for the field, [P] for WSSUS

S_H describes one channel realization. If the channel is wide-sense stationary in t with uncorrelated scattering in τ (WSSUS), then

$$\mathbb{E}[S_H(\tau, \nu) S_H^*(\tau', \nu')] = C_H(\tau, \nu) \delta(\tau - \tau') \delta(\nu - \nu'), \quad (9)$$

and C_H is the *scattering function*. Its marginals are the power–delay profile and the Doppler power profile,

$$P(\tau) = \int C_H(\tau, \nu) d\nu, \quad P_\nu(\nu) = \int C_H(\tau, \nu) d\tau. \quad (10)$$

For a single realization the time-averaged delay profile is $\hat{P}_T(\tau) = \frac{1}{T} \int_0^T |h(t, \tau)|^2 dt$, which converges to P under WSSUS and ergodicity. The wideband analogue replaces (τ, ν) with (τ, α) [5]. Every anchor record states which of these it holds through `statistical_semantics` $\in \{\text{realization, expectation, estimate, posterior_samples, pseudo_label}\}$, and WSSUS, when an operation needs it, is a declared parameter `assume_wssus`, never a default.

Check 2. Are there operational cases where you would want a *local* scattering function (non-WSSUS, per 21) as a first-class kind, rather than a realization plus a time-series of PDPs?

8 Descriptors as functionals of a named representation

The anchor is a vocabulary, not a required payload: every quantity Crossfade estimates is defined as a functional of a *named* channel representation. In the contracts this is one string per quantity, `canonical_kind`, taking one of the values `impulse_response`, `delay_doppler`, `delay_scale`, `power_statistic` or `descriptors_only`, or else the tag `not_channel`. `[G]` that the string exists; `[P]` the list.

8.1 Moments of the power–delay profile

Definition 8.1 (Delay moments). For a non-negative profile P with $0 < \int P < \infty$,

$$\bar{\tau} = \frac{\int \tau P(\tau) d\tau}{\int P(\tau) d\tau}, \quad \sigma_\tau^2 = \frac{\int (\tau - \bar{\tau})^2 P(\tau) d\tau}{\int P(\tau) d\tau}. \quad (11)$$

For a discrete profile $\{(\tau_k, E_k)\}$ with weights $w_k = E_k / \sum_j E_j$: $\bar{\tau} = \sum_k w_k \tau_k$, $\sigma_\tau^2 = \sum_k w_k (\tau_k - \bar{\tau})^2$.

The Doppler spread σ_ν is the same functional of P_ν . Coherence bandwidth and coherence time are of order $1/\sigma_\tau$ and $1/\sigma_\nu$; the constant depends on the correlation level chosen and is a declared convention, not a physical fact.

Definition 8.2 (Resolvable-arrival count). `[P]` Given a delay resolution Δ and a relative threshold $\gamma \in (0, 1)$, cluster the arrivals of a discrete profile greedily in delay order, merging any arrival within Δ of the current cluster; let E_j be cluster energies. Then

$$N_{\text{arr}}(\Delta, \gamma) = \#\{j : E_j \geq \gamma \max_i E_i\}. \quad (12)$$

N_{arr} depends on (Δ, γ) and on the merging rule; two different rules are two different quantities, so the rule is recorded under `assumptions_used` and arrivals inside one cell are listed under `ambiguous`, never silently merged into a point estimate.

8.2 Exact descriptors from an arrival list

For a time-invariant channel given as an arrival list, $h(\tau) = \sum_k a_k \delta(\tau - \tau_k)$, the profile is $P(\tau) = \sum_k |a_k|^2 \delta(\tau - \tau_k)$ and (11) is exact with $E_k = |a_k|^2$. This is why W1’s simulator produces an arrival list: truth is the functional itself, not a second estimate.

Proposition 8.3 (Delay spread from pairwise differences). *With $w_k = |a_k|^2 / \sum_j |a_j|^2$,*

$$\sigma_\tau^2 = \frac{1}{2} \sum_{k,l} w_k w_l (\tau_k - \tau_l)^2. \quad (13)$$

Hence σ_τ is invariant to a common delay shift and is determined by the delay-difference profile $Q(\tau) = \sum_{k,l} |a_k|^2 |a_l|^2 \delta(\tau - (\tau_k - \tau_l))$ through $\sigma_\tau^2 = \frac{1}{2} \int \tau^2 Q / \int Q$.

Proof. Expand $\frac{1}{2} \sum_{k,l} w_k w_l (\tau_k - \tau_l)^2 = \sum_k w_k \tau_k^2 - (\sum_k w_k \tau_k)^2$ using $\sum_k w_k = 1$. $\square$

Remark 8.4 (Why this matters for a passive receiver). A receiver that does not know the source sees $y = s * h$ and can form the autocorrelation $R_{yy}(\tau) = \sum_{k,l} a_k a_l^* R_{ss}(\tau - (\tau_k - \tau_l))$. Its peaks sit at pairwise delay differences with weights $a_k a_l^*$, whose squared magnitudes are $|a_k|^2 |a_l|^2$, exactly the weights of Q . So an autocorrelation-based estimator can recover σ_τ without the absolute delay, *provided* the pairwise differences are resolved ($|\tau_k - \tau_l| > 1/B$ for $k \neq l$) and R_{ss} is narrow relative to them. When differences collide, the estimator is biased; W1 measures that bias against the exact value rather than assuming it away. The absolute delay $\bar{\tau}$ is not recoverable without a source reference, which is the delay ambiguity of Section 11.

Check 3. Do you agree that RMS delay spread (via the difference profile) and resolvable-arrival count (with a declared merging rule) are the right first two descriptors, and that they are what your cepstral and autocorrelation processing already estimate? If you would replace them, which functional of P or C_H , and why?

8.3 The T0 fixture, stated

The W0 workbench's only computation is (11) on a supplied point-weight profile, plus a normalized width $\sigma_\tau/t_{\text{ref}}$. Golden case: delays $\{0, 2\}$ ms, weights $\{1, 1\}$, $t_{\text{ref}} = 2$ ms gives $\bar{\tau} = 1$ ms, $\sigma_\tau = 1$ ms, normalized width 0.5. It exists to exercise units, lineage, replay and refusal, not to estimate anything.

9 Regime coordinates

A regime coordinate is a dimensionless group that a task names as a candidate explanatory variable for a specific descriptor. The fields are **[G]**; whether any group predicts anything is **[H]**.

9.1 Candidate groups

[P]

Group	Definition	Governs	Placeholders (sonar; HF)
M	v/c	Doppler as shift vs. scale	$10^{-3}\text{--}10^{-2}$; $10^{-6}\text{--}10^{-5}$
β_B	B/f_c	sparsity with M , TB , eq. (8)	$0.3\text{--}1$; $10^{-3}\text{--}10^{-2}$
Π_τ	$\sigma_\tau B$	delay spread in resolution cells	$10\text{--}10^3$; $1\text{--}50$
Π_ν	$\sigma_\nu T$	Doppler spread in Doppler cells	$0.1\text{--}10$; $0.1\text{--}10$
Π_L	L/λ	aperture in wavelengths	$10\text{--}100$; $10\text{--}300$
Π_f	f/f_{cut}	propagating-mode count, eq. (16)	$1\text{--}50$; $1\text{--}10$
Π_h	$k h_{\text{rms}} \sin \theta$	coherent vs. diffuse boundary scatter	$0.1\text{--}10$; $0.1\text{--}5$
SNR_{cell}	per-resolution-cell SNR	detection-limited estimation	$-20\text{--}0$ dB; $0\text{--}30$ dB

Ranges are placeholders to be replaced by programme values. Some authors define the Rayleigh roughness parameter with a factor 2; the pack declares which.

9.2 The window question, made dimensionless

The transcript's practical question, what FFT length and window to use for HF versus sonar, has one answer in both domains. For a short-time transform of window length T_w over a total observation T : arrivals separated by less than $1/B$ are not resolved, so the delay structure available to any estimator has Π_τ cells; the window must be short against the coherence time, $\sigma_\nu T_w \lesssim \varepsilon$, while the Doppler structure resolvable over the whole observation has Π_ν cells. Window length follows Π_ν ; resolution follows Π_τ . It is the same decision in both domains once written this way.

9.3 Per-descriptor governing sets and overlap **[P]** for the metric, **[H]** for the matching claim

Each descriptor q names a governing set G_q of groups. A dataset's tested support $C_d(q)$ is a finite set of cells in $\log_{10}$ -coordinates over G_q . Define

$$O_q = C_{\text{src}}(q) \cap C_{\text{tgt}}(q), \quad d_q(c, C_{\text{src}}) = \min_{c' \in C_{\text{src}}(q)} \max_{g \in G_q} |\log_{10} \pi_g(c) - \log_{10} \pi_g(c')|, \quad (14)$$

so d_q is the number of decades a target cell sits from the nearest source cell in the worst governing group. The tested support is the list of evaluated cells, never a bounding box **[G]**.

Descriptor	G_q	Sonar–HF overlap under placeholders
RMS delay spread, arrival count	$\{\Pi_\tau, \Pi_f, \text{SNR}_{\text{cell}}\}$	partial
Interference structure, striation slope	$\{\Pi_f, \Pi_L, \Pi_\tau\}$	partial
Scale or Doppler spread	$\{M, \beta_B, \Pi_\nu\}$	none in M, β_B
Boundary scatter character	$\{\Pi_h\}$	partial

Proposition 9.1 (The overlap result). *Under the placeholder ranges, the sonar and HF intervals for M are separated by at least two decades and those for β_B by at least 1.5 decades. Hence for any descriptor with $M \in G_q$ or $\beta_B \in G_q$, $O_q = \emptyset$ and $d_q \geq 1.5$ for every target cell.*

Corollary 9.2 (Predicted negative control). *[H] If structure transfers only where governing groups overlap, Doppler and scale descriptors are predicted not to transfer between sonar and HF, and whatever transfers lives in $\{\Pi_\tau, \Pi_\nu, \Pi_f, \Pi_L\}$: arrival structure and waveguide interference. This is the transcript’s “ionosphere as the ocean flipped upside down” stated precisely: the shared object is the waveguide, not the Doppler. A measured gain on a Doppler descriptor would be a finding, not a failed gate.*

9.4 The mode-count coordinate

For the Pekeris waveguide of Section 10.1, mode m propagates above its cutoff

$$f_m = \frac{(2m-1)c}{4D\sqrt{1-(c/c_b)^2}}, \quad m = 1, 2, \dots, \quad (15)$$

so with $\Pi_f = f/f_1$ the number of propagating modes is

$$N_{\text{modes}}(f) = \left\lfloor \frac{2fD}{c} \sqrt{1-(c/c_b)^2} + \frac{1}{2} \right\rfloor = \left\lfloor \frac{\Pi_f + 1}{2} \right\rfloor. \quad (16)$$

In the ionosphere the sense is reversed: a hop mode propagates only *below* its maximum usable frequency, $f < f_{\text{MUF}} = f_{\text{crit}} \sec \varphi_0$ (flat earth, no field). A literal f/f_{cut} is therefore not the same quantity in both domains. What is comparable is the propagating-mode count itself, so we propose that each pack define Π_f such that N_{modes} is a monotone function of it, and record N_{modes} alongside.

Check 4. Is the propagating-mode count the right common coordinate for “modes above cutoff” across a fluid waveguide and the Earth–ionosphere duct, or would you define Π_f for HF as f_{MUF}/f , or something else?

Check 5. Does one “medium variability rate” (sound-speed-profile correlation time over T) belong in this list, or do internal waves and fronts need their own coordinate? Please replace the placeholder ranges with programme values where you can.

10 Generators

Both first generators are deliberately simple so that truth is exact by construction. Their limitations are stated; the sonar lead’s acceptance of the first is on the critical path (ADR-11).

10.1 Sonar: image-source Pekeris waveguide

[P]

Isovelocity water column of depth D , sound speed c , density ρ , over a fluid half-space (c_b, ρ_b) [15, 25]. Pressure-release surface at $z = 0$, bottom at $z = D$; source at $(0, z_s)$, receiver at (r, z_r) .

Images. Reflecting the source alternately in the surface ($z \mapsto -z$) and the bottom ($z \mapsto 2D - z$) generates image depths

$$z^{(n,\sigma)} = 2nD + \sigma z_s, \quad n \in \mathbb{Z}, \quad \sigma \in \{+1, -1\}, \quad (17)$$

with surface and bottom reflection counts

$$(n_s, n_b) = \begin{cases} (|n|, |n|) & \sigma = +1, \\ (n-1, n) & \sigma = -1, \quad n \geq 1, \\ (|n|+1, |n|) & \sigma = -1, \quad n \leq 0. \end{cases} \quad (18)$$

(Check: $(n, \sigma) = (0, +)$ is the direct path; $(0, -)$ one surface bounce; $(1, -)$ one bottom bounce; $(1, +)$ surface then bottom.)

Arrival list. For each image, path length, delay and grazing angle are

$$R = \sqrt{r^2 + (z_r - z^{(n,\sigma)})^2}, \quad \tau = R/c, \quad \theta = \arctan \frac{|z_r - z^{(n,\sigma)}|}{r}, \quad (19)$$

and the complex gain, with spherical spreading and a surface coefficient of -1 , is

$$a = \frac{(-1)^{n_s} [R_b(\theta)]^{n_b}}{R}, \quad R_b(\theta) = \frac{\rho_b c_b \sin \theta - \rho c \sin \theta_b}{\rho_b c_b \sin \theta + \rho c \sin \theta_b}, \quad \frac{\cos \theta}{c} = \frac{\cos \theta_b}{c_b}. \quad (20)$$

For $c_b > c$ there is a critical grazing angle $\theta_c = \arccos(c/c_b)$; below it $\sin \theta_b$ is imaginary, $|R_b| = 1$ and R_b carries a phase. That phase is frequency-independent for a lossless half-space, so it is applied to the analytic source signal:

$$y(t) = \Re \left\{ \sum_k a_k s_a(t - \tau_k) \right\} + n(t), \quad h(\tau) = \sum_k a_k \delta(\tau - \tau_k). \quad (21)$$

Truth for W1 is the list $\{(\tau_k, a_k, \theta_k, n_s, n_b)\}$; the descriptors of Section 8 are exact functionals of it. The sum is truncated when τ_k exceeds a declared maximum delay or $|a_k|$ falls below a declared floor; the truncation is recorded as a parameter.

What the model is and is not. With constant reflection coefficients the image sum is the exact solution for the ideal waveguide. With the angle-dependent $R_b(\theta)$ evaluated at each image's specular angle it is the ray (high-frequency) approximation to the Pekeris field: it omits the branch-line (head-wave) contribution and the modal corrections near cutoff, and it has no refraction, so no ducted paths. Mode count is nevertheless well defined through (15), which is how Π_f is assigned to a scene.

Check 6. Is this image model an acceptable first generator for delay-spread and arrival-count descriptors, with a refraction-capable normal-mode or ray code you already run as the second generator for the cross-domain step and the reality gate? Is applying $R_b(\theta)$ beyond the critical angle as a constant phase on the analytic signal how you would do it in the time domain?

10.2 HF: discrete-mode skywave over a Chapman layer [P]

Electron density in a Chapman layer with peak N_m at height h_m and scale height H (overhead sun),

$$N(h) = N_m \exp\left(\frac{1}{2}[1 - \zeta - e^{-\zeta}]\right), \quad \zeta = \frac{h - h_m}{H}, \quad f_p(h) [\text{Hz}] \approx 8.98 \sqrt{N(h) [\text{m}^{-3}]}, \quad (22)$$

critical frequency $f_{\text{crit}} = f_p(h_m)$ [7]. A wave incident on the layer at angle φ_0 from the vertical (flat earth, no magnetic field) reflects where $f \cos \varphi_0 = f_p(h)$, hence only if $f \cos \varphi_0 < f_{\text{crit}}$. Its

group delay is that of the equivalent triangular path reflected at the virtual height (Breit–Tuve and Martyn’s theorems),

$$h'(f_v) = \int_0^{h_r} \frac{dh}{\sqrt{1 - f_p^2(h)/f_v^2}}, \quad f_v = f \cos \varphi_0, \quad \tau_n = \frac{2n}{c} \sqrt{\left(\frac{R_g}{2n}\right)^2 + h'^2}, \quad (23)$$

for an n -hop mode over ground range R_g , with φ_0 found self-consistently from $\tan \varphi_0 = R_g/(2n h')$. Modes are enumerated over n and layers (E, F); the magnetoionic split is modelled as two modes with reflection conditions $f_p = f_v$ (ordinary) and $f_p^2 = f_v(f_v - f_H)$ (extraordinary), f_H the gyrofrequency. Per-mode gain combines spreading, a $1/f^2$ absorption law per D-region passage, and ground reflection on intermediate hops; ionospheric motion is a slow complex multiplicative process per mode with a declared coherence time. Truth is the mode list $\{(\tau_j, g_j(t))\}$, and the received signal has the same form as (21).

10.3 What the two generators share

They share no propagation code and different physics (boundary images in a fluid layer versus refraction in a plasma layer with ground hops). They share one approximation family: *discrete paths with per-path delay and gain*. That family is the anchor assumption itself, and the independence statement required for any cross-domain claim must say so [G]. Measured data is required before any real-world claim.

10.4 The waveguide invariant and its HF analogue

[H]

In a range-independent ocean waveguide the intensity at range r and frequency ω is a sum of modal cross terms $I(r, \omega) \propto \sum_{m,n} A_m A_n^* \cos[(k_m(\omega) - k_n(\omega))r]$. Along a striation the modal phase difference is constant, giving $d\omega/dr = \beta \omega/r$ with

$$\beta = -\frac{\Delta S_p}{\Delta S_g}, \quad S_p = \frac{k}{\omega}, \quad S_g = \frac{dk}{d\omega}, \quad (24)$$

the waveguide invariant [10]; $\beta \approx 1$ for low-order modes in a Pekeris waveguide, and it is estimable without range [6, 14, 26].

For any two discrete paths with delay difference $\Delta\tau(R)$ varying with a range-like parameter R , the interference $\cos[\omega \Delta\tau(R)]$ has striations along $d\omega/dR = -\omega(d\Delta\tau/dR)/\Delta\tau$, so the same construction defines

$$\beta_{\text{path}} = -\left(\frac{d \ln \Delta\tau}{d \ln R}\right)^{-1}, \quad (25)$$

which reduces to (24) for modal pairs. Hypothesis H4 asks whether β_{path} computed from HF hop-mode pairs over ground range is approximately constant across mode pairs and ranges, as β is for low ocean modes; if it is, the striation-slope descriptor is a mechanism that transfers, and if it is not, the analogy breaks at a named place.

Check 7. Is the flipped-ocean analogy strong enough that you would expect (25) to be approximately invariant for HF multimode returns? What would you look for first in a range–frequency display of an HF return?

11 Identifiability

11.1 The generative model

Each domain d is written as

$$y_d = \mathcal{M}_d[\mathcal{F}_d(\theta_{\text{sh}}, \theta_{\text{pr}}; c_d)] + \varepsilon_d, \quad (26)$$

with $\mathcal{F}_d$ the forward (propagation) model, $\mathcal{M}_d$ the measurement and processing chain, c_d the physical context (medium state, geometry, band), θ_{sh} the quantities the task treats as shared and θ_{pr} the rest. “Nuisance” is relative to a task, never intrinsic: each task declares which components of θ are retained, conditioned on, or marginalized $[\mathbf{G}]$. A change to $\mathcal{M}_d$ (resampling, windowing, magnitude-only) is lossy processing; a change to c_d (retune, sound-speed change) is a physical intervention; neither is a “convention”, and each transform a pack admits is classified as exact, coordinate, lossy or physical and tested individually $[\mathbf{G}]$.

11.2 Blind deconvolution

With $y = h * s$ and neither factor known, the following map y to itself:

$$(h, s) \mapsto (\kappa h, s/\kappa), \quad (h, s) \mapsto (h(\cdot - \tau_0), s(\cdot + \tau_0)), \quad (h, s) \mapsto (h * g, g^{-1} * s) \text{ for invertible } g. \quad (27)$$

Scale and delay are the minimal ambiguities; without structure, the whole family of invertible filters is. Identifiability up to scale requires structural constraints on h or s (subspace or sparsity), with conditions on the constraint dimensions relative to the observation length [20]. $[\mathbf{G}]$: a learned estimator does not remove an ambiguity, it hides it, so every estimator lists **assumptions**, **identifiable** and **ambiguous**, and a head that needs an ambiguous quantity returns alternatives or abstains.

11.3 What the classical estimators assume

Cepstrum. For $h = \sum_k a_k \delta(\tau - \tau_k)$ with a dominant first arrival, write $H(f) = a_0 e^{-j2\pi f \tau_0} \prod_{k \geq 1}$ factors; for two arrivals with $r = a_1/a_0$, $|r| < 1$, $\Delta = \tau_1 - \tau_0$,

$$\log(1 + r e^{-j2\pi f \Delta}) = \sum_{n \geq 1} \frac{(-1)^{n+1} r^n}{n} e^{-j2\pi f n \Delta}, \quad (28)$$

so the channel’s complex cepstrum is a train of impulses at quefrequencies $n\Delta$ with weights $(-1)^{n+1} r^n/n$, and $\log |Y| = \log |S| + \log |H|$ separates the source if $\log |S(f)|$ is smooth, meaning its cepstrum lies below the smallest Δ . Assumptions: $|r| < 1$ (the magnitude cepstrum cannot tell the ordering of two arrivals, only $|\Delta|$); arrivals separable in quefrecy ($\Delta > 1/B$); one source. Failure regimes follow: delay collisions, a non-smooth or non-stationary source spectrum, and overlapping sources, for which $Y = S_1 H_1 + S_2 H_2$ does not factor and the logarithm separates nothing.

Artificial time reversal. Beamforming an array toward one ray or mode arrival gives $P_b(f) \approx S(f) G_b(f)$ with $\arg G_b$ approximately linear in f ; then $\arg S(f) \approx \arg P_b(f) - 2\pi f \tau_b + \text{const}$, leaving an overall delay and scale unknown [27, 28]. Assumptions: known array geometry and local sound speed; a resolvable arrival with near-linear phase.

Bilinear models for sources of opportunity. If s lies in a known low-dimensional subspace and h in another, y is bilinear in the two coefficient vectors and the pair is recoverable up to the intrinsic ambiguities under dimension conditions [20, 30]. This is the acoustic template for the overlapped-source problem.

11.4 Multi-view identifiability, and what a deterministic view does not give

Let a recording be $x = f(\mathbf{c}, \mathbf{s})$ with content $\mathbf{c}$ and style $\mathbf{s}$, and let a second view $\tilde{x} = f(\mathbf{c}, \tilde{\mathbf{s}})$ share the content while the style is *independently* resampled. Under the conditions of von K  gelgen et al. [31] and Gresele et al. [11], a contrastive objective on $(x, \tilde{x})$ recovers $\mathbf{c}$ up to an invertible map (block identifiability).

Remark 11.1 (Deterministic re-representations give consistency, not isolation). If $\tilde{x} = \phi(x)$ for a deterministic ϕ (waveform to spectrogram, waveform to estimated CIR), then for any encoder e the pair $(e, e \circ \phi^{-1}|_{\text{range } \phi})$ attains perfect agreement, and no style is stripped because none varies. Such pairs test that an encoder is consistent across representations (what CSI-CLIP++ demonstrates under ideal ray-traced CSI, 16); they are not an alignment basis. Views with independent nuisance variation are: different sub-arrays on one event, different time windows of a stationary source through a slowly varying channel, different upstream chains on one recording, and controlled simulation pairs (same channel, different source; same source, different channel). This is the split between losses L2a and L2b.

11.5 The overlapped-source problem

Many sources in one band, $y = \sum_i h_i * s_i + n$, is blind multichannel separation. It becomes tractable only when the pack supplies structure that makes it identifiable: a source subspace (tonals plus a smooth continuum), the array manifold, and sparsity of each source’s arrival set. An encoder does not discover this; it operates inside it.

Check 8. In the overlapped-source case, what specifically breaks the cepstral method for you: delay collisions, source non-stationarity, or association? Is there a source-subspace assumption that would make it identifiable in the bilinear sense of Tian et al. [30]? Below what Π_τ is arrival sparsity no longer a usable prior?

12 Transfer as a measured quantity

12.1 Losses, budgets, effect

[G]

For a task with descriptor q , an arm a trained with N target labels, and independent units $u \in U$ (scenes, sessions, instruments; never windows), the loss is aggregated at the unit,

$$L_a(N) = \frac{1}{|U|} \sum_{u \in U} \ell(\hat{q}_{a,u}, q_u), \quad \ell = |\hat{\sigma}_\tau - \sigma_\tau| B \text{ for delay spread (error in resolution cells).} \quad (29)$$

The effect of a candidate over a baseline is

$$\delta(N) = L_{\text{base}}(N) - L_{\text{cand}}(N), \quad (30)$$

positive favouring the candidate, with a confidence interval from bootstrap over units and over training seeds. No ratio undefined at zero baseline loss is used.

12.2 Hypotheses

[H]

Hypothesis 12.1 (H1, transfer at overlap). For fixed q and N , on cells in O_q , $\delta(N) = L_{\text{scratch}}(N) - L_{\text{pretrained}}(N)$ exceeds a preregistered meaningful $\delta_{\min}$ with its CI excluding zero.

Hypothesis 12.2 (H2, transfer surface). δ is a decreasing function of $d_q(c, C_{\text{src}})$ in (14). Reported as the surface $\delta(c)$ against d_q ; a gain at large d_q is a finding, never a failed gate.

Hypothesis 12.3 (H3, regime conditioning). Conditioning on regime coordinates improves held-out utility over unconditioned arms under interventions that separate regime from domain identity. Routing telemetry alone is not evidence.

Hypothesis 12.4 (H4, mechanism). β_{path} of (25) estimated in one bounded dispersive waveguide remains predictive in another and beats shuffled-pair and nuisance-matched controls.

Hypothesis 12.5 (H5, uncertainty under shift). Coverage $\text{cov}(c) = \frac{1}{n_c} \sum_{u \in c} \mathbb{1}[q_u \in I_u]$ at nominal level survives shift across cells, or the applicability flag marks the cells where it does not. Reported with interval width, accepted-case error and abstention rate, each with its denominator.

12.3 The baseline ladder as nested comparisons

[G] for the order

Arms in order: (1) classical or direct calculation; (2) physically normalized features plus a small head; (3) independent target model from scratch; (4) target-only self-supervised pretraining; (5) pooled model with a domain token; (6) dense shared/private with adapters; (7) regime-routed experts, optional. Every arm above the first is compared to every arm below it at declared budgets. Arm 3 is mandatory because physics foundation models are conditional generalists with negative transfer in a substantial fraction of cases [4], and arm 7 is an ablation because its motivating result covers two fluid regimes [29]. Well-tuned simple baselines are hard to beat [12].

12.4 Decision rule and what is not evidence

[G]

$\delta_{\min}$ is set from the from-scratch arm’s development-set variability across seeds and frozen with a timestamp before the locked test partition is read. Supported: the CI of δ excludes zero and exceeds $\delta_{\min}$ at two or more budgets. Evidence against: the CI excludes $\delta_{\min}$ in the other direction at two or more budgets. Otherwise inconclusive. A crash is not negative transfer; lack of significance is not evidence of no effect; a result on one simulator relabelled is not cross-domain evidence.

12.5 Fusion, two rules only

[G]

Let $\mathcal{R}(r)$ be the set of root records a result r derives from. If $\mathcal{R}(r_1) \cap \mathcal{R}(r_2) \neq \emptyset$, the two are one piece of evidence and the more conservative (wider) is kept; they are never multiplied. If the model-error dependence between r_1 and r_2 is unknown, they are reported side by side and not combined numerically. Any further rule needs its own association contract, dependence model and calibration test.

Check 9. The pilot is designed so that arrival-structure descriptors have overlapping cells and Doppler descriptors do not, and it measures δ against d_q per descriptor rather than demanding a yes or no. Does that design lose anything you valued in the original conversation? Is descriptor-matched pairing across the two simulators an acceptable primary pairing basis, or do you want pairs defined from a shared physical scene?

Part III

Architecture and contracts

13 Ownership layers

The eight layers below are ownership boundaries, not a mandatory pipeline. A run follows one of three paths (Section 14), chosen deterministically by what the evidence supports. [G] for the ownership rules; [P] for the paths.

Layer	Owns	Guarantee	Must never
L0 Evidence	Immutable artifacts, manifest, lineage	Anything derived resolves to what was observed	Overwrite or reinterpret an artifact
L1 Domain pack	Operations, priors, regime declarations, transforms, tests	Missing capabilities are explicit	Invent a forward model it does not have
L2 Anchor estimators	Pack operations producing anchor estimates	Identifiability stated per quantity	Emit a point where the problem is ambiguous
L3 Encoder	Learned models, when a task allows them	Inputs limited to declared prediction-time fields	See oracle or evaluation-only data
L4 Inference	Heads, forward checks where a forward model exists	Every check has a typed state	Report an unavailable check as passed
L5 Results	Run records with typed states, lineage	A consumer can tell what supports what	Present shared-lineage views as independent
L6 Bench	Splits, leakage checks, reports, claims	Every claim resolves to runs, data, versions	Report a gain without its baseline
L7 LLM	Explanations, pack-metadata drafts	Statements cite records	Write a quantity, uncertainty or verdict

14 Meta-architecture: the simplest thing that meets the guarantees

One Python package, **crossfade**, with five modules; a directory of domain packs that are ordinary Python packages; a store that is a directory of content-hashed files plus a SQLite index; a small CLI. **[P]** for the module boundaries; **[G]** for the constraints that follow.

Constraint	Meaning	Why
No services	Everything runs in one process or one subprocess per numerical worker	Single-user local use is the whole of W0 to W3
No plugin discovery	Packs are listed by import path in crossfade.toml	Discovery is the first thing that silently loads untrusted code; the explicit list is also the trust boundary
No workflow engine	A run is one operation on one record; the “graph” is the list of runs a task produced	A DAG scheduler for a fixture that computes an RMS width is premature framework
No base-class hierarchies	Operation is a Protocol ; packs implement it with plain classes	Inheritance trees are where behaviour hides; a protocol is a checkable shape
No state machines beyond run status	Run.status and the five result-state enums are the only states	The review’s conditional state was a state machine with an authorization step; an assumption is a parameter
Storage is files plus SQLite	Payloads at store/blobs/<sha256> ; records, runs, reports as JSON files and rows	Immutable by construction: there is no update API
Five record types	Record , TaskSpec , Operation , Run , Report	The review’s thirteen carry the same information
Physics lives in packs	The core never branches on a domain, a representation kind or a regime coordinate	The core is a ledger and a bench; packs are the science

Modules. `records` (the `Record` type, manifest, hashing, store: xarray to zarr for arrays, JSON for structured records, SQLite index); `tasks` (`TaskSpec`, versioned Pydantic models loaded from YAML); `packs` (the `Operation` protocol with `assess` and `run`; explicit registry); `runs` (`Run`, execution, replay); `bench` (splits, leakage checker, reports, claims; arrives in W1).

Three execution paths, one ledger. Figure 1 shows the whole of it. A run follows one path, chosen by `Operation.assess`, never by a planner and never by an LLM:

- supported, op has no model → *reference path*: classical op → `Run`;
- supported, op is a model → *learned path*: model op → `Run`, inputs limited to prediction-time fields;
- record is a summary → *output-only path*: `import_summary` → `Run` with `upstream_asserted` quantities;
- unsupported or unknown → no run; the assessment is returned with reasons and missing items.

All three write the same `Run`; the bench reads runs and writes reports; nothing else exists in the core.

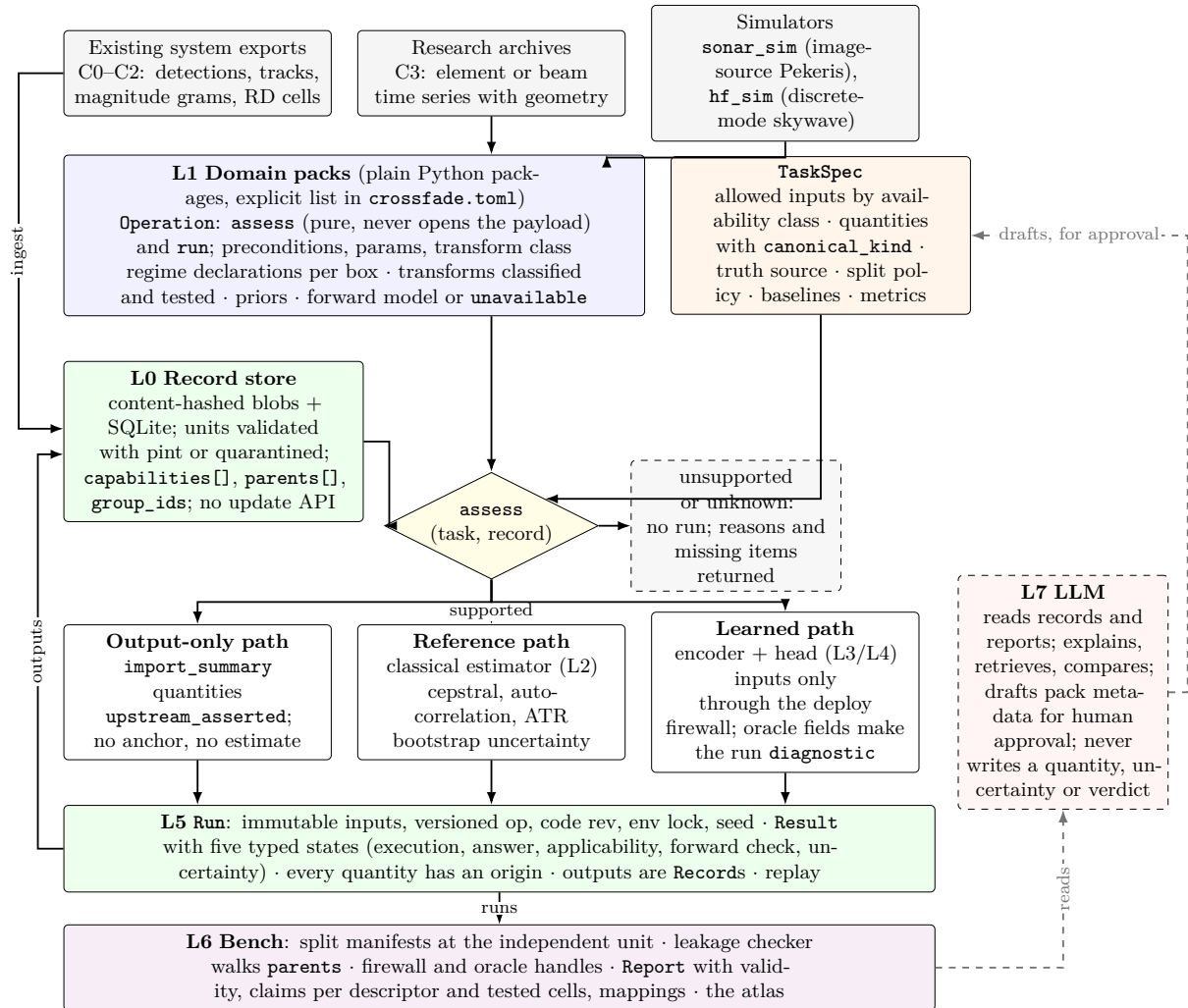


Figure 1: The architecture. Inputs at three capability levels enter through domain packs (L1) into an immutable record store (L0). For each task, `assess` reads the record’s declared capabilities and the task’s requirements and chooses one of three paths, or refuses with reasons. All three paths write the same `Run` (L5) with five typed result states; the bench (L6) reads runs and writes reports and claims. The LLM (L7) reads records and reports and drafts pack metadata for human approval; it never writes a number. Physics lives entirely in the packs; the core is a ledger and a bench.

Size budgets. Budgets enforce the constraints. Core at end of W0: 1500 lines (review at 2500). Fixture packs: 300. `sonar_sim` including the simulator: 1200. `hf_sim`: 1000. `bench.py`: 500. W2 models: 2000. Any single function: 60. An overrun is not a failure; it is a mandatory review that asks what became a framework and whether it needed to.

What was rejected, and where it came from. Regime-routed mixture-of-experts as the architecture (spec v0.1); a mandatory full-grid anchor payload from every pack (v0.1); universal PINN losses (transcript framing); the LLM inside the numerical path (transcript); thirteen record types with a `conditional` state (review); twelve slices with uncertainty as its own slice (review); “no shared code” replaced by documented independence (review; we kept both); calendar commitments removed entirely (review; we kept labelled planning estimates); xarray attributes as the unit system (v0.1); transforms as conventions (v0.1); deterministic paired views as an identifiability basis (v0.1); a `Mapping` written by every experiment (v0.1); H2 as a gate (v0.1); microservices, a graph database, a GNN evidence graph and a model-registry service (plan 2, v0.1).

15 The five record types

All are Pydantic v2 models serialized to JSON. Unknown fields are rejected on load, so a contract change is a version bump, never a silent addition. Field lists are **[P]**; the invariants in Section 16 are **[G]**.

15.1 Record

Anything observed or derived: observation, representation, anchor estimate or structured summary, one type distinguished by `kind`. Key fields: `id`, `kind`, `payload_kind` (array or structured), `payload_ref`, `byte_hash`, `manifest_hash`, `source_system`, `acquired_at` (may be unknown, never inferred), `ingested_at`, `data_use`, `capability_label` (C0–C3, a summary only), `capabilities[]` (explicit: `has_phase`, `has_array_geometry`, `has_source_reference`, `has_calibration`, `has_timing`, `upstream_asserted`), `coords`, `units`, `masks`, `group_ids` (session, instrument, environment, source recording, scene; used by splits), `processing_history[]` (each step with op id, version, params and transform class), `parents[]`, `produced_by_run`, `validation_state`, `validation_notes[]`, `regime` (per coordinate: value, units, provenance, availability, uncertainty), and `anchor` (required for anchor estimates: kind, descriptors each with definition and canonical kind, statistical semantics, normalization, assumptions, identifiable, ambiguous, uncertainty reference).

Identity. `id` derives from the byte hash and the manifest hash together; deriving it from the manifest alone would collide for two payloads with one manifest. The manifest covers units, coordinates, masks, processing definitions and interpretation version, so identical bytes labelled seconds and milliseconds are different records. `produced_by_run` is provenance, not identity, so a replay that reproduces identical bytes and manifest reuses the record. Correcting metadata creates a successor with a parent link; the original is never edited.

Payloads and quarantine. Arrays are xarray datasets with named dimensions and coordinate variables, stored as zarr. Units are validated at ingest with pint; an attribute is a label, not a check. Any unit failure, or any data variable without units, stores the record with `validation_state = quarantined`; it is listable and refused by every assessment until a successor supplies what is missing.

Capabilities. The C0–C3 labels remain as a summary vocabulary and never authorize anything. C0: upstream summaries (detections, tracks, scalar features; `upstream_asserted`). C1: magnitude-only time–frequency views (`has_timing`). C2: complex coefficients, beam-level STFT or range–Doppler cells (`has_phase`, `has_timing`). C3: element or beam time series with geometry. A record labelled C3 without a source reference still fails the preflight of an operation that needs one.

Lineage. `parents[]` records derivation only, in the spirit of PROV-O without importing the ontology. Shared parents mean shared evidence and nothing more is inferred from lineage.

15.2 TaskSpec

What a run may see and must return, loaded from YAML with the version in the filename: `task_id`, `version`, `task_kind` (transform, estimation, comparison), `question`, `allowed_inputs[]` each with an availability class (`prediction_metadata`, `prediction_evidence`, `prediction_estimate`, `training_only`, `evaluation_only`), `required_capabilities`, `params` (each with `type`, `units`, `required`, `is_assumption`), `quantities` (each with `definition`, `units`, `canonical_kind` or `not_channel`, `ambiguity_allowed`), `truth_source`, `normalization` (rule, reversibility, fit scope), `regime_coords[]`, `baselines[]`, `metrics[]`, `split_policy`, `uncertainty_requirement`, `completion_rule`, `promotion_rule`.

The firewall lives here: the deploy handle returns only fields whose availability is one of the first three classes; training-only and evaluation-only fields are served through a different function, and any run that receives them is tagged `diagnostic`. Assumptions are parameters: a physical assumption an operation needs is a parameter with `is_assumption = true`; if absent, `assess` returns `unsupported` naming it. Nothing is ever assumed on the caller’s behalf.

15.3 Operation

A pack’s executable unit, a protocol: `op_id`, `version`, `preconditions[]` (capabilities the record must carry), `params_model` (unknown params refused), `transform_class` (exact, coordinate, lossy, physical, none), `determinism` (exact or tolerance), `is_model`, and two methods. `assess(task, record, params)` is pure and fast: it reads capabilities, validation state, units, coordinates, processing history and params, never the payload, and returns `supported`, `unsupported` or `unknown` with named reasons and missing items. Value-level checks (a non-finite unmasked sample, zero total weight) therefore happen in `run` and produce `answer = unsupported` with reasons in diagnostics while the run itself succeeds. The registry reads the explicit pack list, imports each module, checks `CORE_API_VERSION == 1`, and fails whole on any error; there is no partial registry.

15.4 Run

One execution attempt: `run_id`, `task` and `op` with versions, `inputs[]`, `params`, `assumptions_used`, `code_rev`, `env_lock_hash`, `seed`, `path` (reference, learned, output_only, diagnostic), timestamps, `status` (succeeded, failed, cancelled), `attempt_of`, `error`, and a `result` holding `quantities` (each with `value`, `units` and `origin`: `upstream_asserted`, `computed`, `inferred`), `outputs[]`, and the typed states of Section 20. Publication order: output records first, then the run JSON, then the SQLite row, each via temp file and atomic rename; a crash between steps leaves orphan outputs that `verify` reports, never a run referencing missing outputs. `replay` re-resolves the op by id and version, re-executes with the same inputs, params and seed, compares by the determinism contract, and records a new run with `attempt_of` set.

15.5 Report

One declared comparison: an **ExperimentSpec** (task, data manifests with roles, arms with exposure, primary metric, effect rule with a frozen timestamp, seeds, tuning policy, budgets, **exploratory** flag), a **SplitManifest** (unit, partitions, frozen timestamp, hash), **runs[]**, per-arm metrics with intervals and denominators, observed budgets, **validity** (valid, invalid, incomplete), deviations, **claims[]** and **mappings[]**. A **Claim** names direction, descriptor, the regime cells actually tested, arm and baseline arm, outcome (supported, evidence_against, inconclusive), effect with interval, and limitations. A **Mapping** is recorded only for an implemented correspondence. Claims are written only from valid reports; a failed arm is a failed run and never evidence against anything.

16 Invariants

Each has a test; the slice column names where it is first enforced. All [G].

Id	Invariant	Slice
CF-01	Payload blobs are written once and never modified; there is no update path	W0
CF-02	Record identity covers payload bytes and semantic manifest; same bytes with different units are different records	W0
CF-03	Every array record has validated units on every data variable and coordinate, or is quarantined	W0
CF-04	A capability label never authorizes an operation; assess checks explicit capabilities and params	W0
CF-05	assess returns unknown when prerequisites cannot be established; unknown is never coerced to supported	W0
CF-06	A task fixes allowed inputs by availability class; the deploy handle cannot return training-only or evaluation-only fields	W0
CF-07	An anchor estimate may omit its payload; no field is fabricated to satisfy a shape	W0
CF-08	A deterministic transform reports uncertainty not_applicable ; a learned estimate without validated uncertainty reports unavailable ; neither is ever a number	W0
CF-09	A forward check with no forward model is unavailable , never passed	W0
CF-10	Every run resolves to immutable inputs, a versioned op, a code revision and an environment lock	W0
CF-11	Failed and cancelled runs are persisted and linked; a retry is a new run with attempt_of	W0
CF-12	Outputs are published before the run that references them	W0
CF-13	Every quantity carries an origin: upstream_asserted , computed or inferred	W0
CF-14	Splits are at the declared independent unit; every derived record inherits its root's group ids; the checker refuses a partition crossing	W1
CF-15	Oracle-provenance regime coordinates and evaluation-only fields never reach a run on the learned or reference path; a run that uses them is diagnostic	W1
CF-16	Test membership is frozen before any pretraining; unlabeled test exposure is recorded as exposure	W2
CF-17	A valid negative or inconclusive report completes an experiment; an invalid or incomplete report produces no claim	W1
CF-18	Every claim names its arm, baseline arm, direction, descriptor, regime cells tested, effect and budgets	W1
CF-19	Tested support is the list of evaluated cells; the atlas never marks an untested interior cell validated	W1
CF-20	Shared-parent results are one piece of evidence; unknown model dependence withholds numerical fusion	W3
CF-21	No pack operation writes to an upstream system	W3

Id	Invariant	Slice
CF-22	Anchor-estimate uncertainty is propagated or the result is tagged <code>conditional_on_plugin</code> naming the estimate	W1
CF-23	Every quantity in a task names a <code>canonical_kind</code> or is tagged <code>not_channel</code>	W0
CF-24	Every transform in a pack has a class and an equivalence test	W1
CF-25	Unknown fields on any record type are rejected on load	W0

17 Domain packs

A domain pack is a plain Python package exposing a manifest and operations. The core registers packs from an explicit list; the platform owns execution, records and evaluation; the pack owns domain meaning. **[G]** for the boundary; **[P]** for the element list: **manifest** (name, version, core API version, operation ids, fixtures), **operations**[] (at least one), **regime** (declared ranges per group, placeholder or measured; canonical anchor kind per box), **transforms** (every transformation applied or admitted, classified and tested), **tests** (golden fixtures, known failure cases, regime boundaries where a method stops working), and optionally **priors**, **forward_model** (declared **unavailable** when absent), **anchor_estimators** and **vocabulary**.

Transform classification. Under the generative model (26):

Class	Examples	Expected behaviour	Test
Exact conversion	Unit relabel, sample-index origin, complex sign convention	Invariant	Output identical after inverse conversion
Coordinate transform	Array rotation, beam steering, reference-frame change	Equivariant	Output transforms by the declared rule
Lossy processing	Resampling, window length and overlap, magnitude-only, decimation, CFAR thresholding	Declared information loss; applicability may change	Retained-information contract stated and checked on fixtures
Physical intervention	Carrier retune, sound-speed change, ionospheric state change, source motion	Answer may change	Not an invariance; recorded as a context change

Passive sonar pack, sketch. Operations: beamform-and-STFT (lossy, parameters recorded), cepstral arrival estimation, ray-based blind deconvolution, striation-slope estimation, later a learned descriptor estimator, each declaring preconditions (array geometry, sound speed at the array, band). Regime: M up to 10^{-2} , β_B up to 1, Π_f 1 to 50, placeholders. Canonical kind: **delay_scale** when Π_s exceeds the declared threshold, else **delay_doppler**. Priors: sparse ray arrivals with surface and bottom images; approximately linear modal phase in frequency for low-order modes; waveguide-invariant striation slope. Forward model: the image-source generator of Section 10.1; a normal-mode or ray model later; a neural-operator surrogate of a parabolic-equation solver is a later option [18].

HF skywave pack, sketch. Operations: matched filter and Doppler processing (lossy), multimode delay-Doppler estimation from range-Doppler cells, sparse CIR reconstruction under interference. Regime: M around 10^{-6} , β_B around 10^{-2} , Π_ν set by ionospheric coherence time. Canonical kind: **delay_doppler**. Priors: few discrete modes with ordinary and extraordinary splitting; slow multiplicative fading; mode delay ordering. Forward model: the discrete-mode generator of Section 10.2; a published ray tracer once its license and callable form are verified.

Output-only pack. A legacy system connects with one operation, `import_summary`, no forward model and no anchor estimator. It participates in comparison and lineage warnings, and every request for reconstruction returns `unsupported` with the missing capabilities named. This is how existing programmes join without touching their processing chains, and it ships independently of any transfer result.

Part IV

Estimators, models and the training objective

18 Anchor estimators

Every anchor estimator states what it can and cannot recover from its inputs (Section 11). A capability label never authorizes an estimator; its own `assess` checks its preconditions and refuses with the missing items named.

Estimator	Level	Recovers	Assumes	Ambiguous
Striation slope, waveguide invariant β	C1	Range-rate to range ratio, mode interference structure	Few-mode waveguide, broadband source	Absolute range without a frequency anchor
Cepstral multipath	C1/C2	Relative arrival delays, boundary reflection structure	Dominant first arrival, arrivals separable in quefrequency	Arrival ordering; delays within one cell
Artificial or synthetic time reversal	C3	Source waveform and source-to-array impulse response	Linear modal phase, known geometry and local sound speed	Overall delay and scale
Ray-based blind deconvolution	C3	Impulse response and source waveform, broadband	Resolvable ray arrivals, known geometry	Source spectrum absolute level
Bilinear models for sources of opportunity	C3	Channel from unknown but structured sources	Source in a known low-dimensional subspace	Source-channel split within the subspace
Wideband ambiguity, Mellin-domain scale	C2/C3	Scale α and delay τ jointly	Known or estimated source, wideband regime	Sign of scale for symmetric spectra
Delay-Doppler multi-mode	C2	Mode delays and Dopplers	Narrowband regime, few discrete modes	Ordinary-extraordinary assignment
Sparse CIR reconstruction under interference	C2	Full-band CIR from partial spectrum	Sparse arrivals, temporal smoothness	Arrivals within one resolution cell

Learned estimators. Trained on the pack’s forward model with the same structural priors as losses (sparsity in delay, smoothness in time, forward consistency through $\mathcal{M}_d$), following the physics-informed transformer precedent that reached power-delay-profile similarity 0.82 against 0.62 for the best baseline at fifty percent spectrum occupancy [33]. A learned estimator inherits the classical estimator’s assumption list and declares its training regime box. An estimated anchor used as a training target is a `pseudo_label` and the record says so.

Estimator output. An anchor record: a kind, an optional payload, named descriptors as functionals of the canonical kind, statistical semantics, and the three lists **assumptions**, **identifiable**, **ambiguous**. A downstream head that needs an ambiguous quantity returns a distribution over the alternatives or abstains.

Out of scope here. Detection, classification and tracking heads sit on top of results and are owned by each programme’s existing systems or by later task-specific work.

19 Models

The first learned candidate is a dense shared/private encoder with domain adapters [3], trained only after the baseline ladder of Section 12 has been run. Regime-routed mixture of experts is an ablation arm, not the architecture. **[P]** for the candidate; **[H]** for every claim that one arm beats another.

Why routing is demoted. The 2026 result that motivated it covers two fluid regimes and reports domain-separated routing with low validation error [29]. That is evidence routing can avoid gradient conflict between incompatible regimes; it is not evidence that routing is best for inverse channel tasks, and a router fed regime coordinates that trivially predict the target is not discovering physics. The 2026 benchmark showing physics foundation models to be conditional generalists with negative transfer in a substantial fraction of cases [4] is why the from-scratch arm is mandatory.

Losses. Each is individually validated in the literature; their composition is the hypothesis.

Loss	Form	Views or inputs	Claim it supports
L1 Masked modelling	Reconstruct masked patches of the native view	Unlabeled archives per domain	Label-free pretraining, as in masked spectrogram modelling for radio and the RF foundation models
L2a Representation consistency	Contrastive between deterministic representations of one recording	Same recording, deterministic transforms	Encoder consistency across representations [16]; not identifiability
L2b Content isolation	Contrastive between views with independent nuisance variation	Sub-arrays of one event; time windows of a stationary source; different upstream chains; controlled simulation pairs	Isolation of shared physical content up to an invertible map [11, 31]; the alignment basis
L3 Cross-domain anchor alignment	Contrastive between shared latents of records from different domains whose canonical-kind descriptors match, restricted to the descriptor’s overlapping regime box	Descriptor-matched pseudo-pairs, tagged as such	Sharing where physics says it may exist; never whole-domain alignment [32]
L4 Structural physics	Arrival sparsity in τ , temporal smoothness of h , forward consistency where $\mathcal{F}_d$ exists	Anchor estimates	The physics-informed part that applies to channels [27, 33]
L5 Transform consistency	Invariance under exact conventions only; equivariance under coordinate transforms; tested behaviour under lossy processing	Pack transform declarations	The transform classification
L6 Private preservation	Reconstruct native view from shared and private latents; orthogonality penalty	All	Domain detail retained [3]

Pretraining plan. L1 begins once the split policy and transform contracts are frozen and test membership is fixed; “unlabeled” does not exempt data from exposure rules. L2a and L4 follow once anchor estimates exist. L2b requires paired views from the pairing protocol. L3 runs only at declared regime overlap per descriptor. L5 and L6 throughout. Task heads last.

Budgets. Adaptation efficiency (target labels, target unlabeled exposure, adaptation compute) and total cost (simulation, pretraining, tuning, inference) are two separate views, always both. For any sparse arm, active and total parameters are both reported.

Not the objective. Latent overlap, embedding visualizations, and domain indistinguishability under a discriminator are not success criteria.

What transfers today, in these domains. ImageNet-pretrained vision models slightly beat audio-pretrained networks on passive sonar target recognition: what transferred was spectrogram texture, not acoustics [23]. Masked-autoencoder pretraining on communications I/Q transferred to radar signal recognition with a one-shot gain close to in-domain pretraining [13]. CSI-CLIP++ transferred an RF channel encoder across three tasks and across simulators under ideal ray-traced

CSI [16]. The pattern: shallow transfer from generic image pretraining is easy and weak; deeper transfer appears when the objective is built on paired physical views of the channel. The domain-expert tier is being built by others (RF foundation models; an underwater acoustic foundation model under contract). The differentiated contribution is the anchor, the regime atlas and the evidence layer.

Part V

Inference, uncertainty and typed states

20 Typed result states

Every result carries five independent state fields. No single Boolean stands in for them, and no state is ever filled by default with a passing value. [G]

Dimension	States	Rule
Execution	succeeded, failed, cancelled	A failed run publishes no quantities
Answer	complete, partial, abstained, unsupported	partial names what is missing
Applicability	in_scope, out_of_scope, unknown	unknown is the default, never in_scope
Forward check	passed, failed, unavailable, not_applicable	Absent forward model means unavailable, never passed
Uncertainty	posterior, confidence_interval, predictive_interval, ensemble_summary, unavailable, not_applicable	A deterministic transform is not_applicable; a learned estimate without validated uncertainty is unavailable

Each quantity is tagged `upstream_asserted`, `computed` or `inferred`, so a consumer can tell a legacy system’s own confidence from a Crossfade estimate.

21 Inference methods

Not one algorithm. A deterministic transform returns a value. A classical estimator returns a value and whatever error characterization its literature supports; in W1, a block bootstrap over time segments. Where a pack has a forward model and a prior, simulation-based inference is the preferred route to a posterior: train a neural posterior estimator on $(\theta, \mathcal{M}_d[\mathcal{F}_d(\theta)] + \varepsilon)$ pairs and amortize on real observations, using the `sbi` library before anything custom, with the diagnostics the guides prescribe [8, 9]. SBI is a method a pack may use, not a return type the platform demands.

Forward validation. Where $\mathcal{F}_d$ and $\mathcal{M}_d$ exist, the engine predicts what the pack would observe under the inferred explanation and compares at the observation level actually available. A check on data that trained the estimator is labelled in-sample. Agreement is necessary and not sufficient; when the posterior is multimodal or an ambiguous quantity is involved, `alternatives[]` is populated.

Uncertainty record. Beyond its kind, an uncertainty carries the estimand, the conditioning information, the method and version, and the population and regime box on which it was validated. A result that plugs in a point anchor estimate says `conditional_on_plugin` and

names the estimate; otherwise the anchor’s own uncertainty is propagated. Post-hoc calibration degrades under shift [24] and conformal guarantees hold under exchangeability or specific departures, so applicability is reported alongside every uncertainty and **unknown** is a legitimate value.

Abstention. A head may abstain when applicability is out of scope or unknown, or when a forward check fails a declared threshold. Abstentions are runs with **answer = abstained**, so the abstention rate is a metric with a denominator.

Fusion. The two rules of Section 12 and nothing more until a use case exists.

Part VI

Evaluation protocol

22 Three levels of demonstration

Level	Shows	Does not show
E0 Contract fixture	T0 computes a known property of a supplied profile; storage, units, capability handling, replay work	Anything about channels or transfer
E1 Within-domain estimation	A declared descriptor is estimated from permitted observations and compared to independent truth with its own error model; classical and from-scratch baselines characterized	Transfer
E2 Cross-domain transfer	Only the declared transfer intervention changes; target utility measured under held-out conditions; shared-simulator smoke tests, independent-domain evidence and measured-data validation reported separately	Universal generalization

An E0 result is never promoted to E1 or E2 by relabelling data.

23 Preregistration, firewall, splits, pairing

Preregistration. An experiment is confirmatory only if every field is filled: task version, data manifests, roles, arms with exposure, primary metric, effect rule with a frozen timestamp, seeds, tuning policy, budgets, independent unit, expected failure modes, exclusions. A spec missing any of these may run as exploratory and produces no claim. No universal percentage threshold is invented; baseline variability is estimated on development data, the meaningful effect is agreed, and the rule is frozen before the locked test is viewed.

Feature and truth firewall. Every input field has one availability class. The deploy handle serves only prediction-time fields. Oracle fields are served by a separate function and any run that calls it is diagnostic and cannot satisfy the promotion gate. Anchor estimates used for pairing are pseudo-labels carrying estimator version, lineage and uncertainty. Test reference answers never choose routing, positive pairs or normalizers on the deploy path. In simulation

this is the main leakage risk: a coordinate computed from the simulator’s true delay spread, feeding a model whose target is delay spread, is the answer in disguise.

Splits and exposure. At the task’s independent unit: session, environment, instrument, scene or source recording. Every derived record inherits its root’s group ids; the checker walks parents to the root and refuses a partition crossing. Test membership is frozen and hashed before any pretraining. Target-domain unlabeled exposure is recorded per arm. Normalization, regime selection, pairing thresholds and calibration are fit on training or training-plus-development partitions only.

Pairing. Each pair record declares one basis: `same_event` (two instruments observed one physical event), `controlled_mechanism` (two independent simulators rendered a deliberately corresponding mechanism), `descriptor_matched` (pseudo-pair by matching estimated descriptors; weakest), or `negative`. They are never pooled. Shuffled-pair and nuisance-matched controls run where the task’s shortcut risks call for them.

Simulator independence. A cross-domain claim from simulation requires both no shared forward-physics kernel and a written independence statement covering scene distribution, approximation family and truth construction. Code independence is the cheap auditable proxy; assumption independence is the target; neither substitutes for the other or for measured-data validation.

24 Gates

Gate	Question	Passes when	Does not imply
Delivery	Did the declared arms run validly and reproducibly?	Required arms present; leakage checks pass; report rebuilds from persisted runs	Any scientific result
Scientific	What does the result support?	A scoped claim with effect, interval, budgets and limitations, in any direction	Deployment fitness
Promotion	Is the model useful for its declared inputs?	Preregistered utility and uncertainty criteria met at the independent unit on relevant data	Anything beyond tested cells

An experiment is complete when a preregistered comparison ran validly and produced a report. A model is promoted when the report supports its declared utility. A hypothesis is supported, contradicted, or inconclusive. None of the three implies another.

Part VII

Build plan

25 Slices

Four slices ship now and three wait for a reason to exist. Each is a complete user-visible operation with a demo and a test suite. Estimates are planning numbers for the funding conversation, labelled as such, and assume the sonar lead’s W1 input arrives in the first two weeks. [P]

Slice		Deliverable	Gate	Estimate	Blocked by
W0	Work-bench	Ingest a supplied profile; compute T0; preserve units and lineage; replay; refuse reconstruction on a summary record	Delivery	2–3 weeks, one engineer	Nothing
W1	First task	Simulated beam time series from the image-source waveguide; RMS delay spread and arrival count via cepstral and autocorrelation estimators; bootstrap uncertainty; bench with grouped splits and leakage checks; a report characterizing the classical estimators	Delivery and scientific	4–6 weeks, two people	Sonar lead confirms the quantity and the simulator
W2	Transfer charter	HF pack with an independent multimode simulator; SSL arm; dense shared/private arm; one preregistered transfer trial per descriptor with typed uncertainty	Scientific, any direction	8–12 weeks, two to three people	W1 split policy frozen; pairing basis; independence statement
W3	Output-only adapter	Read-only import of one existing system’s summary export; comparison; duplicate-evidence warnings; no upstream write	Delivery	2–3 weeks, parallel to W1	Which system and export format
Later:	routing	Dense versus routed ablation	After W2 shows a dense arm worth beating	Deferred	W2 result
Later:	explanations	LLM over records; pack-metadata drafts for approval	After W1 records are stable	Deferred	Stable Run and Report
Later:	fusion	Numerical combination under a declared dependence model	After a use case exists	Deferred	A use case

Dependencies. W0 and W3 share **Record** and **Run** and nothing else. W1 shapes the contracts alongside W0 and starts the day its quantity is confirmed. W2 depends on W1’s split policy for its pretraining start and on W1’s estimators for its baselines.

What each gate refuses. W0: any anonymous tensor; any record without explicit capabilities; any fabricated default for a missing unit, reference scale or convention; any assessment that coerces unknown to supported. W1: any learned-arm result without ladder arms 1 and 2; any random-window split; any oracle coordinate on the reference or learned path; any estimator without a typed uncertainty or an explicit unavailable. W2: any gain measured on one simulator relabelled; any missing independence statement; any pretraining before test membership is frozen; any calibration claim validated only on the training cells; any claim from an invalid or incomplete report. W3: any write to upstream processing; any duplicate evidence counted twice; any upstream confidence relabelled as a Crossfade estimate.

Team. One person on packs and estimators (the sonar lead’s domain), one on the core and bench, and from W2 one on models.

26 W0: the workbench, as built

Implemented test-first on branch `w0-workbench`: five core modules (**records**, **tasks**, **packs**, **runs**, **db**) plus the CLI, two fixture packs, two task specifications, the golden fixture, and 39 tests

mapped one-to-one to the PRD’s acceptance tests; ruff clean; core at 804 lines against a 1500-line budget. Three details settled during implementation are reflected in the contracts: value-level checks live in `run`, not `assess`; `Record.id` derives from both hashes; `produced_by_run` and `validation_notes` were added. The worker runs in-process; subprocess isolation with a resource limit is the next increment and changes no contract. The branch is not merged pending the sonar lead’s review of the architecture.

27 W1: the first real task

Task specifications. `pdp.rms_delay_spread` and `pdp.arrival_count`, both estimation tasks. Allowed inputs: beam time series (prediction evidence); sampling rate, band, array geometry, local sound speed (prediction metadata); the arrival list (evaluation only); simulator parameters (training only). Required capabilities: `has_timing`, `has_phase`. Parameters: window length (lossy, recorded), cepstral lifter, `assume_point_source` (an assumption). Quantities: RMS delay spread in seconds and arrival count, both with canonical kind `power_statistic`, the count with ambiguity allowed. Truth: exact functionals of the arrival list. Regime coordinates: Π_τ (inference, estimated; and oracle for diagnostics), Π_f (inference, from band and depth if supplied), SNR_{cell} (estimated). Split policy: unit is the scene; held-out groups are scene and environment. Uncertainty: ensemble summary or predictive interval, never unavailable.

Regime grid. Water depth $\{50, 100, 200\}$ m; range $\{1, 3, 10\}$ km; bottom $\{\text{sand, silt, rock}\}$ as sound speed and density pairs; centre frequency $\{200, 500, 1000\}$ Hz with fractional bandwidth $\{0.3, 0.6\}$; SNR per cell $\{-10, 0, 10\}$ dB: 486 cells, three scenes per cell with different source and receiver depths. Cells are the atlas’s tested support.

Operations. `beamform_stft` (lossy; declares the window length and ships a test that delay spread from two window lengths agrees within a declared tolerance on the golden scene); `autocorr_pdp` (anchor kind `power_statistic`, PDP on a delay grid plus the two descriptors; block bootstrap); `cepstral_pdp` (kind `descriptors_only`; block bootstrap); `direct_from_truth` (diagnostic path). Each estimator lists `identifiable = [rms_delay_spread]` and `ambiguous = [arrival_count within one cell]`.

Experiment W1-E1. Arms: the two estimators and the diagnostic sanity arm. Primary metric: absolute error in RMS delay spread in resolution cells, per scene. Secondary: arrival-count error under the stated merging rule; bootstrap coverage at the nominal level; abstention rate below a declared SNR floor. Effect rule: characterization, not superiority; per-cell error and coverage tables. Seeds: five. Claims: one per estimator per descriptor, supported if coverage is within tolerance across tested cells, else evidence against with the failing cells listed. These are claims about the estimator’s error model, not about transfer. Definition of done: ten acceptance tests green, a report in the store, and a one-page summary for the sonar lead of which cells each estimator is trustworthy in.

28 W2: the transfer charter

Software. `packs/hf_sim` with the discrete-mode simulator, an `import_rangedoppler` lossy transform, a `multimode_dd` estimator producing a `delay_doppler` anchor with the two descriptors, and a bootstrap uncertainty; an independence statement stored as a record; model operations in each pack for ladder arms 2 to 6, all reading inputs through the deploy firewall; pairing records with a declared basis (primary `controlled_mechanism`: for each cell in the overlap the two simulators render scenes whose PDP descriptors are deliberately matched; secondary

`descriptor_matched`, never pooled); the regime coverage report produced before any transfer run.

Arms and budgets.

Arm	Ladder	Pretrained from	Target labels	Target unlabeled
A1 classical	1	none	0	0
A2 features + head	2	none	N	0
A3 scratch	3	none	N	0
A4 target SSL	4	target unlabeled	N	all train
A5 pooled	5	both, labeled	N	all train, both
A6 dense shared/private, source-pretrained	6	source SSL then source labeled	N	source all; target 0
A6' dense shared/private, source + target SSL	6	source and target SSL	N	all train, both

Learning curve $N \in \{8, 32, 128, 512\}$ target scenes. Controlled: target labels, adaptation compute, tuning budget. Reported: upstream pretraining compute, total parameters. Seeds: five. Independent unit: scene, held out by scene and environment.

Effect rule, to be frozen before the locked test. Primary metric as in W1. Effect $\delta(N) = L_{A3}(N) - L_{A6}(N)$. Meaningful $\delta_{\min}$: the smallest delta exceeding two seed standard deviations of A3 on the development set at $N = 32$, frozen and timestamped before the test partition is read. Decision as in Section 12.

Shortcut controls. Shuffled-pair control (A6 with pairing bases permuted; a gain surviving this is not transfer); nuisance-matched control (target scenes matched on SNR and bandwidth only); oracle-regime diagnostic (A6 given oracle Π_τ ; diagnostic path; cannot promote). The predicted negative control is a Doppler-spread descriptor added to both packs if time allows.

Open inputs that block the scientific gate, not the software. Pairing-basis approval (sonar lead); HF simulator simplification acceptance (an HF-familiar reviewer); whether the W1 error model is tight enough in the overlap cells to serve as pseudo-label truth for A6' (numerical reviewer); a named measured dataset for the reality check (sonar lead).

29 W3: the output-only adapter

One pack with two operations: `import_summary` parses an existing system's export into summary records with origin `upstream_asserted`, the system's confidence kept in diagnostics and never under uncertainty, and group ids from whatever session or file identifiers the export carries; `compare_summaries` aligns two summary records on time and declared keys, lists agreements and disagreements, and warns when both resolve to the same parents. Non-goals: any estimate, any anchor, any learned op, any numerical fusion, any write to the upstream system, any live connection. A static check confirms no file writes outside the store. Blocked only by which system and which export format (ADR-25).

30 Data and simulators

Nothing is invented as available. The two in-house generators are specified in Sections 10.1 and 10.2. Public candidates for the reality gate, each with what must be verified before use:

SWellEx-96 (current hosting and terms; which arrays and events are public; source-track resolution); SBCEX17 ship-noise recordings (whether the recordings or only derived products are public); DeepMIMO (simulated, so it serves the independent-generator check, not the reality gate); a measured indoor or vehicular time-varying CIR dataset with enough sessions for grouped splits (to be identified). Programme data enters only at W3 through the output-only adapter unless C3 archives are already cleared for research use; every record carries the owner’s `data_use` label; nothing assumes a new collection campaign. Every dataset is described by a manifest record (source, license or clearance, capabilities, group-id scheme, truth kind and error model, regime coverage as a list of cells, known gaps) before any run consumes it, and the bench produces from the manifests the per-descriptor regime coverage and intersection before any transfer hypothesis runs.

31 Decision register

Id	Decision and choice	Tag	Status	Blocks
ADR-01	Product unit: capability-aware task execution with an auditable ledger; transfer is a measured extension	P	Adopted	–
ADR-02	Five record types	P	Adopted	–
ADR-03	Anchor as a typed family with optional payload; descriptors as functionals of a canonical kind; canonical kind per regime box	P	Adopted	–
ADR-04	Three execution paths chosen by assess; no planner, no DAG engine	P	Adopted	–
ADR-05	Explicit capabilities plus per-op preconditions; C0–C3 as labels only	G	Adopted	–
ADR-06	Assumptions are parameters; no conditional state	G	Adopted	–
ADR-07	Content-hashed files plus SQLite; zarr for arrays; JSON for structured	P	Adopted	–
ADR-08	pint validation at ingest; quarantine on failure	G	Adopted	–
ADR-09	Explicit pack list; one core API version integer	G	Adopted	–
ADR-10	First scientific task: RMS delay spread and arrival count from beam time series, simulated, C3	P	Proposed, awaiting sonar lead	W1
ADR-11	W1 simulator: in-house image-source Pekeris waveguide; refraction-capable code as second generator later	P	Proposed, awaiting sonar lead	W1
ADR-12	Truth exact from the arrival list in W1; estimated anchors are pseudo-labels everywhere	G	Adopted	–
ADR-13	Regime coordinates task-scoped, provenance-bearing, nullable, with availability; per-descriptor matching	G / H	Adopted	–
ADR-14	First learned arm: dense shared/private with adapters after ladder arms 1–5; routing optional	P	Adopted	W2
ADR-15	SSL starts when W1’s split policy is frozen and test membership hashed	G	Adopted	W2
ADR-16	Typed uncertainty kinds; block bootstrap for classical estimators in W1	G / P	Adopted	–
ADR-17	Forward checks only where a pack declares a forward model	G	Adopted	–
ADR-18	Report holds claims; Mapping only for implemented correspondences; tested cells listed	G	Adopted	–
ADR-19	Simulator independence: no shared kernel and a written statement, both	G	Adopted	W2

Id	Decision and choice	Tag	Status	Blocks
ADR-20	Output-only adapter proceeds without any transfer result	G	Adopted	–
ADR-21	Two fusion rules only	G	Adopted	–
ADR-22	LLM reads records, drafts for approval, never a quantity or verdict	G	Adopted	–
ADR-23	Adaptation and total cost reported separately; active and total parameters for sparse arms	G	Adopted	–
ADR-24	Named datasets with fields and truth before any real-world claim	G	Open	W2 reality gate
ADR-25	W3 target system and export format	P	Open	W3
ADR-26	Shared deployment only when concurrency exists	P	Open, nonblocking	–
ADR-27	Size budgets; overrun triggers review	G	Adopted	–
ADR-28	Dependency gates for sequencing; labelled planning estimates	P	Adopted	–

Before an estimator becomes a reference operation, the sonar lead and a numerical reviewer approve, per representation: its definition, sign and normalization conventions, statistical meaning, physical approximations, and its conformance tests.

Part VIII

The language model, and the people

32 Where the LLM sits

The LLM never touches a number. It reads typed records, each estimate with its assumptions, its uncertainty kind and the cells it was validated on, and it explains, retrieves, compares hypotheses and says what evidence is missing. It has no authority to write a quantity, an uncertainty, or a verdict on whether a mapping is valid. [G]

Its one genuinely new job is upstream of the models: turning a conversation like the original one into a draft domain pack, a structured list of assumptions, admissible transforms, known failure regimes and test cases, for a human to approve. Operator feedback splits three ways: presentation (“this explanation was confusing”), metadata (“this run used a different window”), and scientific adjudication (“independent evidence says this estimate was wrong”). Only the third is a training label, and only after it is versioned. Fine-tuning on operator feedback comes after the evidence layer can tell those three apart, not before.

33 Open questions for the sonar lead

The first block is on the critical path; W1 cannot be specified without it. The rest can arrive over the first month.

The first real task (blocks W1).

- Is RMS delay spread plus resolvable-arrival count from beam time series the right first descriptor? If not, which functional of the PDP or scattering function, and why?
- Which propagation code should generate truth for it, and what are that code’s known accuracy limits, in dB and in which regimes?

- Which classical estimators for it do you already run, at which capability level, and what are their known failure regimes?
- What metadata is genuinely available at inference time for a real recording: array geometry, local sound speed, band, nothing else?

Anchor and regime.

- Do the programme ranges for M , β_B , Π_τ , Π_ν , Π_f and per-cell SNR match the placeholders? Replace them.
- Is delay-scale the right canonical kind for the passive wideband case, or does the modal picture carry structure that delay-scale loses at the ranges you care about?
- Does one medium-variability rate capture what matters, or do internal waves and fronts need their own coordinate?
- For HF: is the flipped-ocean analogy strong enough that a waveguide-invariant-like descriptor should exist in multimode returns? What would you look for first?

Estimators and identifiability.

- In the overlapped-source case, what specifically breaks the cepstral method: delay collisions, source non-stationarity, or association?
- Is there a source-subspace assumption that would make the overlapped-source problem identifiable in the bilinear sense?
- Below what Π_τ is arrival sparsity not a usable prior?

Data and scope.

- Which archives exist at C3 and could be cleared for research pretraining? Volume and span?
- Which public ocean-acoustic datasets do you trust for the reality gate, and which environmental truth do they carry?
- Is there any paired data, one event seen by an acoustic and an RF system, or is cross-domain pairing purely descriptor-matched?
- Should the first HF pack target skywave OTH, or a simpler narrowband RF channel with richer public data, with OTH second?
- Is a C0 output-only adapter for the current operator tool the right W3 target, or is there a C2 system that gives a stronger first integration?

On the composition.

- The design composes the first plan’s inference formulation with the second plan’s workbench and the review’s contracts. Does the composition lose anything you valued in either?
- Unlabeled pretraining starts as soon as splits are frozen, ahead of any benchmark result. Do you see a risk in that?

34 The pitch, in one paragraph

We are building a reusable inference layer that connects existing sensing systems through a common physical anchor: the channel’s spreading function, which every wave-propagation domain already estimates in its own way. Each domain keeps its own models and processing. The platform records what each system observed, under which assumptions, and with what uncertainty, and it tests where learned structure actually transfers across domains at matched physical regime. The immediate deliverable is consistent integration and traceable, uncertainty-aware analysis. The research upside is that each new domain or environment needs less bespoke data and modelling, because the parts of the physics that are shared are learned once.

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